Problem Statement 4

HITACHI

Find-it-by-Action (FIBA AI)

Short Description

FIBA AI is a zero-shot, edge-friendly system that enables users to search videos using natural language actions such as “cutting an onion” or “opening a box” in egocentric (first-person) videos. Instead of relying on heavy action recognition models, it detects the object being manipulated, tracks its interaction with human hands, and infers actions using lightweight motion patterns such as rotation, translation, and expansion. The system extracts and highlights key moments in the video where the queried action occurs, providing an efficient and explainable approach to video understanding that can run on edge devices (e.g., mobile phones).

Resource Availability

- MediaPipe → Real-time hand detection
- WeDetect→ Detect objects from text
- EdgeTAM→ Segment and track objects
- SANSA: SAM2 Few-Shot Segmentation
- Dataset: https://cbs.ic.gatech.edu/fpv/

(EGTEA Gaze+ raw videos from dropbox)

Objectives

Develop a zero-shot, edge-friendly video understanding system that:

- Takes a video + text query as input
- Identifies the active object being manipulated
- Tracks it across frames
- Infers the action using motion logic (not heavy ML)
- Extracts key moments where the action occurs

Deliverables / Outcome

- Working demo (video + query → result)
- Visual output:
- ✓ Bounding box / mask on active object
- ✓ Key frames or short clips
- Brief logic explanation:
- ✓ Detected rotation → inferred opening”
- Optional UI (Gradio/Web app)

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